

CO-2-AT2 - Class Test

- 1) Data mining:- Data mining is the Process of extracting useful Pattern and Knowledge from large databases

Two Data-mining tasks :- 1) Classification
2) Clustering.

KDD Process:- 1) Data Cleaning
2) Data integration
3) Data selection
4) Data Transformation

- 5) Data mining
6) Pattern Evaluation
7) Knowledge Presentation

8A
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- * Removes errors and missing values
- * Improves data quality and mining accuracy.

- b) Data cleaning:- The process of removing incorrect, incomplete or inconsistent data.

→ Two Data quality issues:- * Missing values, * Noisy data

→ Handling missing values:- * Replace with mean/median.
* Ignore the record.

→ Handling Noisy Data:- * Binning, * Regression.

- 2) Data Integration:- Data integration combines data from multiple sources into one data set.

Two challenges:- * Data redundancy.
* Data inconsistency.

→ Handling Redundancy:- Remove duplicate records.

→ Handling Inconsistency:-

- * Standardize data formats and values.

b) Covariance Analysis:

Covariance measures how two variables change together.

Two Covariance Formulas:

1) Population Covariance :-
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

2) Sample Covariance :-
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3)
a)

Stocks A and B.

Covariance = +5 (Positive).

* The Stock Prices rise together.

* They are positively correlated and not independent.

b) Data Mining System Architecture.

Components:-

- * Data Sources
- * Database / Data Warehouse
- * Data Cleaning & Integration
- * Database Server
- * Data Mining Engine
 - * Pattern Evaluation.
 - * Knowledge Base
 - * User Interface (GUI)

Role:- These components work together to discover useful knowledge from data.

Covariance:

Given,

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 1\}$$

$$\text{Covariance} = -0.45$$

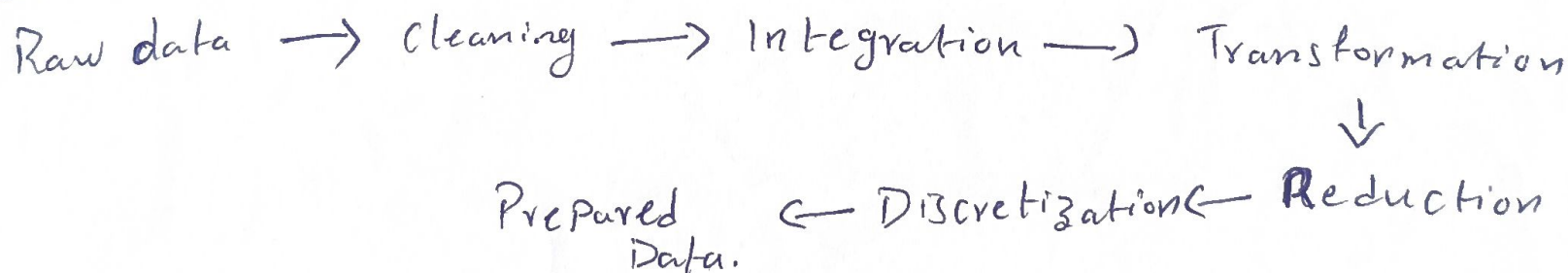
Interpretation: Negative Covariance means that 'Variables moves in opposite directions.'

Data Preprocessing:

Major steps:-

- 1) Data cleaning
- 2) Data Integration
- 3) Data Transformation
- 4) Data Reduction
- 5) Data Discretization

Pipeline:



Equal-Frequency Binning: Equal frequency binning divides data into bins with an equal number of values.

For 27 values, using 3 bins:

* Bin 1: 13 - 22 (9 values)

* Bin 2: 22 - 35 (9 values)

* Bin 3: 35 - 70 (9 values).

b) Data Discretization:- Data discretization Converts Continuous data into intervals.

Importances:-

- * Reduces noise.
- * Simplifies analysis
- * Improves mining Performance.

Equal-width	Equal-frequency
* Equal interval / width * Simple to implement * Sensitive to outliers	* Equal number of records * Better for uneven data. * Handles skewed data better